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PSAC Exam Practice · Predicted Specimen Paper

2026

SCIENCE

Grade 5 · Paper 4

Habitats and Adaptation | Changes of Materials | The Water Cycle

Time Allowed: **1 hour**

Total Marks: **50**

Name: _____

Class: _____ Date: _____

INSTRUCTIONS TO CANDIDATES

1. Write your name, class and date in the spaces provided above.
2. Answer **ALL** questions in **all three sections**.
3. Write your answers clearly in the spaces provided.
4. The mark for each question or part question is shown in square brackets [].
5. Diagrams should be clear and neatly labelled.

Section	Topic	Available	Score
A	Living Things — Habitats and Adaptation	20	
B	Matter and Materials — Changes of Materials	15	
C	Energy and Forces — The Water Cycle	15	
TOTAL		50	

SECTION A — Living Things: Habitats and Adaptation [20 marks]

1 Match each animal to its habitat by drawing lines.

[4]

Animal		Habitat
Camel	• •	Pond / freshwater lake
Polar bear	• •	Desert
Frog	• •	Arctic / icy regions
Toucan	• •	Tropical rainforest

2 A camel is well-adapted to live in the desert. Study the list of adaptations and answer the questions below.

[5]

- Hump stores fat as a food reserve
- Wide, flat feet spread weight on sand
- Long eyelashes protect eyes from sand and dust
- Can survive long periods without water
- Thick fur insulates against cold desert nights

(a) Name **two** features of a camel that help it survive without food or water for a long time.

[2]

(b) How do wide, flat feet help the camel in the desert?

[1]

(c) The desert can be very cold at night. Name **one** adaptation that helps the camel survive the cold.

[1]

(d) What is the meaning of the word *adaptation*?

[1]

3 Tick **ONE** box to show the correct answer.

[4]

(a) A habitat is:

- ☐ the food an animal eats
- ☐ the natural place where an organism lives
- ☐ the group of animals in a forest
- ☐ the body parts of an animal

(b) Which adaptation helps a polar bear survive in the Arctic?

- ☐ thin, dark fur to absorb heat
- ☐ thick white fur for insulation and camouflage
- ☐ long legs to run fast across sand
- ☐ webbed feet for climbing trees

(c) A fish has streamlined body shape to help it:

- ☐ attract a mate
- ☐ breathe on land
- ☐ move easily through water
- ☐ camouflage itself on land

(d) If a habitat is destroyed, what is most likely to happen to the organisms that live there?

- ☐ they will grow larger
- ☐ their population will increase
- ☐ they will die or be forced to move
- ☐ nothing will change

4 Write **TRUE** or **FALSE** for each statement.

[3]

Statement	True or False
Desert plants often have spines instead of leaves to reduce water loss.	
All animals in a rainforest live on the ground.	
Animals can adapt to their habitat over many generations.	

5 A cactus plant lives in the desert. Describe **two** adaptations of the cactus that help it survive with very little water.

[4]

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SECTION B — Matter and Materials: Changes of Materials [15 marks]

- 6 Sort the following changes into **reversible** or **irreversible** by writing each one in the correct column. [6]

burning wood cooking an egg dissolving salt freezing water
melting chocolate rusting iron

Reversible Change	Irreversible Change

[1 mark per change correctly placed]

- 7 Tick **ONE** box to show the correct answer. [4]

(a) Which of the following is an example of an *irreversible* change?

☐ melting ice ☐ freezing juice ☐ burning paper

☐ dissolving sugar

(b) When iron rusts, a new substance is formed. This means rusting is:

☐ a reversible change ☐ a physical change

☐ an irreversible chemical change ☐ a change of state

(c) Baking a cake is an irreversible change because:

☐ it gets hot

☐ the raw ingredients cannot be recovered from the baked cake

☐ it changes colour ☐ it uses electricity

(d) Which of the following is evidence that a chemical change has taken place?

☐ the material changes shape

☐ a new substance with different properties is formed

☐ the material is crushed ☐ the material is dissolved in water

- 8 A student mixes vinegar and bicarbonate of soda in a beaker. Bubbles form and the beaker gets cold.

(a) Name the gas produced when bubbles form.

[1]

(b) Give **two** pieces of evidence that show a chemical change has taken place.

[2]

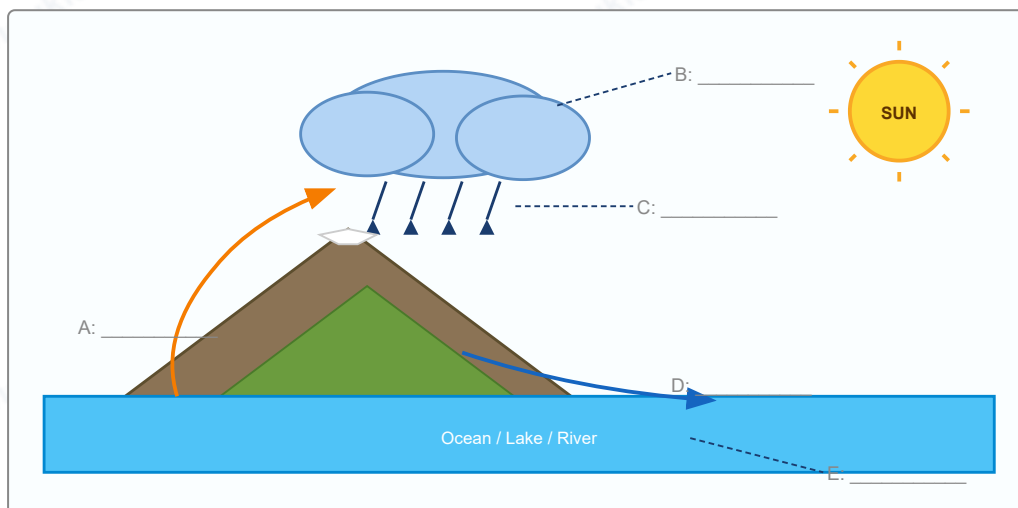
(c) Is this change reversible or irreversible? Give one reason.

[2]

SECTION C — Energy and Forces: The Water Cycle [15 marks]

- 9 The diagram below shows the water cycle. Label the stages A–E using the words in the box. [5]

condensation evaporation precipitation run-off collection



- 10 Answer these questions about the water cycle. [5]

(a) What provides the energy that drives the water cycle? [1]

(b) What change of state takes place when water evaporates from the ocean? [1]

(c) Why does water vapour form clouds when it rises into the atmosphere? [1]

(d) Name **two** forms of precipitation. [2]

- 11 Tick **TRUE** or **FALSE** for each statement about the water cycle. [3]

[3]

Statement	True	False
Water is continually recycled in the water cycle — no water is created or destroyed.	☐	☐
Water evaporates only from the ocean, not from land or plants.	☐	☐
Condensation occurs when water vapour cools and turns back into liquid water droplets.	☐	☐

12 Tick **ONE** box to show the correct answer.

[2]

(a) In the water cycle, water vapour rises because:

- ☐ it is heavier than air
- ☐ warm air carries it upward
- ☐ the moon pulls it up
- ☐ gravity pulls it upward

(b) Which human activity can disrupt the water cycle by reducing the amount of water that reaches rivers?

- ☐ planting more trees
- ☐ deforestation (cutting down trees)
- ☐ building more schools
- ☐ swimming in lakes

END OF PAPER 4 · Total Marks: 50

MARK SCHEME — Grade 5 Science · Paper 4

For teacher / parent use only. Accept any reasonable scientific answer that conveys the same meaning. Award full marks where a candidate's response is scientifically correct even if worded differently. Do not penalise poor spelling unless it changes the scientific meaning.

SECTION A — Habitats and Adaptation [20 marks]

Q	Part	Acceptable Answer	Marks
1	—	Camel → Desert · Polar bear → Arctic/icy regions · Frog → Pond/freshwater lake · Toucan → Tropical rainforest (1 mark each)	4
2	(a)	Hump stores fat as a food reserve [1] AND can survive long periods without water [1] (any 2 from the list)	2
	(b)	Wide, flat feet spread the camel's weight so it does not sink into the sand.	1
	(c)	Thick fur (insulates against cold desert nights)	1
	(d)	A feature (or special characteristic) that helps an organism survive in its environment. Accept: a special feature that suits an animal/plant to where it lives.	1
3	(a)	the natural place where an organism lives	1
	(b)	thick white fur for insulation and camouflage	1
	(c)	move easily through water	1
	(d)	they will die or be forced to move	1
4	—	TRUE / FALSE / TRUE (1 mark each)	3
5	—	Any two correct adaptations, 2 marks each: thick waxy stem stores water [1] + explanation of how it helps [1]; spines instead of leaves reduce water loss [1] + explanation [1]; long/deep roots to absorb water [1] + explanation [1]. Award 2 marks per well-described adaptation (max 4).	4

SECTION B — Changes of Materials [15 marks]

Q	Part	Acceptable Answer	Marks
6	—	Reversible: dissolving salt, freezing water, melting chocolate · Irreversible: burning wood, cooking an egg, rusting iron (1 mark each correctly placed)	6
7	(a)	burning paper	1
	(b)	an irreversible chemical change	1
	(c)	the raw ingredients cannot be recovered from the baked cake	1
	(d)	a new substance with different properties is formed	1

Q	Part	Acceptable Answer	Marks
8	(a)	Carbon dioxide	1
	(b)	Any two: bubbles formed (gas produced) [1], temperature dropped/beaker got cold [1], new substance formed [1]	2
	(c)	Irreversible [1] — because new substances are produced that cannot be changed back into vinegar and bicarbonate of soda [1]	2

SECTION C — The Water Cycle [15 marks]

Q	Part	Acceptable Answer	Marks
9	A–E	A: evaporation · B: condensation · C: precipitation · D: run-off · E: collection (1 mark each)	5
10	(a)	The Sun (solar energy)	1
	(b)	Evaporation (liquid → gas)	1
	(c)	As water vapour rises, it cools. Cooling causes the water vapour to condense (change back to tiny liquid droplets) which form clouds.	1
	(d)	Any two: rain, snow, sleet, hail (1 mark each)	2
11	—	TRUE / FALSE / TRUE (1 mark each)	3
12	(a)	warm air carries it upward	1
	(b)	deforestation (cutting down trees)	1